

IPA ASSIGNMENT – 17

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1. Store one RGB image using 3 2d matrices

```
from PIL import Image
import numpy as np
image = Image.open("img_2.jpeg").convert("RGB")
img_array = np.array(image)
R = img_array[:, :, 0]
G = img_array[:, :, 1]
B = img_array[:, :, 2]
print("Red Matrix:")
print(R)
print("\nGreen Matrix:")
print(G)
print("\nBlue Matrix:")
print(B)
print("\nImage Size:", image.size)
```

OUTPUT

```
Red Matrix:
[[ 27  18   6 ... 215 208 230]
 [ 59  40  15 ... 217 210 231]
 [ 53  37  13 ... 219 210 231]
 ...
 [236 237 237 ... 159 160 159]
 [245 245 246 ... 176 177 176]
 [180 179 179 ... 190 191 190]]

Green Matrix:
[[ 39  30  15 ... 205 198 220]
 [ 71  52  23 ... 207 200 221]
 [ 64  49  21 ... 209 200 221]
 ...
 [247 248 248 ... 179 180 179]
 [255 255 255 ... 195 196 195]
 [191 190 190 ... 207 208 207]]

Blue Matrix:
[[ 39  30  14 ... 196 189 211]
 [ 71  52  25 ... 198 191 212]
 [ 66  49  23 ... 200 191 212]
 ...
 [251 252 252 ... 190 191 190]
 [255 255 255 ... 202 203 202]
 [197 196 196 ... 214 215 214]]

Image Size: (800, 520)
```

2. Implement rgb to grayscale of the image

```
import cv2
import matplotlib.pyplot as plt
image = cv2.imread("img_2.jpeg")
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original RGB Image")
plt.axis("off")
plt.subplot(1, 2, 2)
plt.imshow(gray_image, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")
plt.show()
cv2.imwrite("grayscale_image.jpg", gray_image)
```

OUTPUT

Original RGB Image



Grayscale Image

